Graduation Thesis Draft

MSc in Security and Network Engineering (SNE)

Introduction & Literature Review

Topic [Draft]: Multi-Layered AI-Augmented Defensive Phishing System

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Abstract

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Chapter 1

Introduction

1.1 Phishing as an Intensifying Cybersecurity Challenge

This paper will provide an extensive literature survey of the current state of phishing email detection systems, and propose multiple multi-layered defensive architectures in the Era of AI-Boosted Phishing with a specific emphasis on quantum-enhanced techniques. Although phishing detection systems have been extensively researched, NLP-centric approaches, such as semantic and deep learning-based models, are underexplored, fragmented, and not adequately benchmarked. The paper argues that phishing emails remain a persistent cybersecurity threat and that traditional defensive countermeasures are actively exploited and bypassed. It remains the most economically expensive form of cybercrime (APWG). It will also address the attacker's techniques used to bypass these measures. Despite decades of defensive academic research, it remains the most successful attack vector for attackers [1], [2].

Phishing is defined as a social engineering-driven offensive vector that deceives computer users into disclosing sensitive information such as: credentials, banking data, and identity details. Mostly by email (or could be SMS, QR, Voice, Video) as the most common attack surface due to its scalability, low operational cost, and high trust inheritance within organizational workflows [2]. The surface of email attacks has witnessed exponential growth, especially during crises (e.g., COVID-19, Financial crises, wars, APTs, Political Events, Business Investments, Banking Strategies, State-Sponsored APTs), due to the rapid exploitation of fear, uncertainty, and institutional disruption (APWG reports cited in [2]).

The persistence of phishing is economically significant: dealing with a single phishing incident requires considerable labor, response time, and remediation costs, which scale even more dramatically at the enterprise level [1]. As emphasized in the literature, phishing is not a static threat but an adaptive ecosystem that continuously evolves to bypass deployed defenses [3], [2].

The data provided by the literature review reveal a serious mismatch in organizational resource allocation. On average, when an organization spends 27.5 minutes and $31.32 to mitigate a single danger, it engages in asymmetric war of attrition. In this model, the attacker's marginal cost of sending an additional 10,000 emails is nearly nil, whereas the defender's costs increase with each successful perimeter bypass [4]. Furthermore, the increasing challenge is the move from broad phishing campaigns to spear phishing and whaling, in which AI-driven social engineering (Generative AI) enables adversaries to circumvent traditional mail filtering, such as poor language and suspicious links, at scale.

The dual nature of phishing:

- Technical deception (malicious links, malware, spoofed URLS, attach-

ments, Embedded HTML)

- Psychological manipulation (urgency, fear, authority, trust, spear phishing, whaling)

Chapter 2

Literature Review

A major challenge identified is the lack of a universally accepted definition of phishing [2], with current definitions varying in:

1. Target (credentials vs. broader attacker benefit)
2. Attack medium (email, websites, malware)
3. Attacker impersonation scope (third parties vs. direct deception)

The dual-nature character of phishing is reflected in this definitional diversity. First, it incorporates technical deception such as: malicious URLs, spoofed domains, malware attachments, embedded scripts, and HTML obfuscation. Second, it makes extensive use of psychological manipulation, urgency, fear, authority, trust, and personalization. These include spear-phishing and whaling campaigns [2], [5]. Therefore, effective defenses should handle both technical artifacts and linguistic intent, which many systems fall short in this regard.

2.0.1 History of Phishing

The history and evolution of phishing dates back to 1995, when the first recorded use of social engineering was utilized to target AOL accounts [1], [5]. It got evlolved from basic credential theft to more sophisticated attacks such as: e-commerce impersonation, online fraud, and Man-in-the-Browser (MitB) attacks. Modern phishing now targets individual users, employees, organizations, vendors, supply chain, and service providers, often using personalized data within the enterprise domain (spear-phishing, whaling).

The literature review categorizes phishing detection systems into three broad families:

Blacklist-Based Detection	Traditional Machine Learning	Deep Learning
Relies on known malicious senders, URLs, or domains.	Uses classifiers such as: SVM, Random Forest, Decision Trees, Logistic Regression, Naïve Bayes	Introduced to override feature engineering limitations
Strengths: simple, fast	Manual feature engineering: Email body features, Subject line patterns, URLs and scripts, Sender metadata	Uses CNNs, RNNs, RCNNs, word embeddings, attention mechanisms.
Reactive rather than proactive	Key limitations: High dependence on surface-level text, Poor generalization to paraphrased/semantically altered emails, Labor-intensive (dataset, LLM fine-tuning) and brittle feature design	Advantages: Automatic feature extraction, Better modeling of context and structure.

Requires manual reporting	-	Limitations: Large data requirements, Training & fine-tuning complexity and costs, Sensitivity to missing email components (e.g., headers), Architecture complexity
Ineffective against zero-day phishing attacks	-	-

2.0.2 NLP-Specific Features for Phishing Detection

TABLE 2.1
Major Feature Categories [2]

Parameter	**Value**
Email body features	HTML structure, suspicious phrases, embedded links.
Subject line features	Urgency cues (“verify”, “account”, “debit”), reply indicators
URL-based features	IP-based URLs, “@” symbols, redirections, external/internal link ratios.
Script-based features	JavaScript, pop-ups, on-click actions.
Sender-based features	Mismatch between sender address and reply-to address.

2.0.3 NLP and ML for Phishing Email Detection

Summarized literature on NLP and ML for phishing email detection [paraphrase]

Early NLP-based phishing detection relied on [6]:

- Bag-of-Words
- N-grams
- Markov chains
- Topic models

Key problems identified:

- Over-reliance on surface text
- Inability to detect semantic rewording or intent manipulation
- Vulnerability to minor linguistic changes

Recent advances include [7], [8]:

- Word embeddings (e.g., Doc2Vec)
- Hybrid feature selection methods
- Multi-stage ML pipelines
- Deep learning architectures with attention

Some DL-based models report very high accuracy (>99%) but [2]:

- Often evaluated on limited or non-standard datasets
- Lack generalization to real-world, evolving phishing tactics
- Fail in specific edge cases (e.g., headerless emails, short URLs)

2.0.4 Key Gaps and Limitations Identified

The literature review carefully identified several research gaps and challenges in email phishing detection systems. A significant factor that contributed to this limitation is the lack of a genuine solution that could provide the semantic understanding of email's intent. As many systems rely on the shallow hard-coded lexical or structurual cues rather than the deeper contextual interpretations.

This reliance is compounded by the extensive focus on URLs and attachmenets, often at the expense of detection phishing emails that rely solely on persuasive or deceptive language. Existing approaches also demonstrate poor performance when faced with highly personlaized spear-phishing attacks, which deliberately evade the generic pattern detection systems and exploit the organizational trust relationships [1], [2].

In addition, the absence of a standardized benchmark datasets with an updated corpus data by the open-source community remains a significant barrier to fair comparison, reproducibility, and results accuracy across studies. Many proposed models are evlauted based on small, outdated datasets, limiting their validity and assessment on real-world email corpora. Furthermore, human factors - such as adversary motivation, biases, background, and psychology are minimally considered in most technical solutions. Collectively, these shortcomings highlight the need for defensive solutions that integrate semantic intent with human-centric considerations in order to produce a reliable phishing countermeasure [1], [9].

2.0.5 Discussion and Research Direction

The paper will emphasize a solution that should remediate the identified gaps using a Multi-Layered Computational Defensive Architecture that can be deployed at various points in an email solution using state-of-the-art systems design [prp]. The solution will include various semantic NLP and intent analysis models with examination, benchmarking, and fine-tuning findings. These shall include:

- ML Models
  - Logistic Regression, Linear SVM, Multinomial NB, Random Forest, Gradient Boosting, SGD Classifier, Passive Agressive, AdaBoost,

Ridge Classifier, Complement NB, Extra Trees, XGBoost, LightGBM.

  - FastAPI

- AI Models

  - NLP (BERT, RoBERTa)
  - LLMs (Qwen3 4B, DeepSeek-R1, Mistrall-Small-24B-Insruct)
  - LLM + Vision (Qwen3-VL-Embedding-8B, Qwen3-Embedding-8B, Qwen3-VL-Reranker-8B, Qwen3-VL-8B-Instruct)
  - Quantum NLP

- Bypass Techniques for Email Protection System

  - Spear Phishing, Whaling, GenAI Phishing, LLM Layer Prompt Injection, LLM Denial of Service, LLM Supply Chain Vulnerabilities, LLM Manipulation, LLM Data Leakage

- Conclusion

  - LLM + Vision -> Image/HTML Based Phishing
  - Quantum-Enhanced AI

- Technology Stack

  - 7 DevSecOps Pipelines
  - git, GitHub, Docker
  - HashiCorp Vault
  - GitHub Actions, Jenkins, Gitlab

- Azure DevOps
- NPM, NodeJS
- Kubernetes
- Gatekeeper, Kyverno, OPA
- Prometheus + Grafana
- Compliance & Supply-Chain Integrity: SLSA Level 3+
- Tekton
- Bazel
- SBOM: Syft
- Grype
- in-toto
- Sigstore / Cosign
- SAST, DAST
- Terraform, Ansible, Bicep
- Code Signing: Windows Authenticode
- OCI-compliant registry
- Postfix: Mail Transfer Agent (MTA)
- Dovecot: POP3, IMAP, and Managesieve server
- Nginx: Web Server
- OpenLDAP, ldapd: LDAP server, used for storing mail accounts
- MySQL, MariaDB, PostgreSQL: Store application data

- mlmmj: Mailing list manager
- Amavisd-new: Interface between Postifx and SpamAssassin, ClamAV
- SpamAssassin: Content-based Spam Scanner
- postgrey: greylisting
- ClamAV, VirusTotal: Virus scanner
- Roundcube webmail: Webmail (PHP)
- SOGo Groupware: provide calendar (CalDAV), contact (CardDAV), tasks and ActiveSync
- z-push: Exchange ActiveSync
- Fail2ban: Scan log files and bans IPs that show the malicious signs
- iRedAPD: postifx policy server with SRS (Sender Rewrite Scheme)
- Nextcloud: CardDav/CalDAV
- DNSSEC,NSD4, SPF, TLS, DKIM, & DMARC: DNS Security
- MTA-STS, SSHFP: Mail Client Security
- duplicity: Backups
- munin: System monitoring

2.0.6 Quantum Paradigms

There are several implementations of Quantum-Enhanced Algorithms for Natural Language Processing in academia for pattern recognition that aid in enhancing the performance of phishing detection, such as PhishVQC, and in certain cases outperforming classical computing ML models, such as QSVMs

[10]. The paper will explore several implementations and the underlying theory in model optimization techniques.

Bibliography cited

[1] M. Khonji, Y. Iraqi, and A. Jones, "Phishing detection: A literature survey," *IEEE Communications Surveys amp; Tutorials*, vol. 15, no. 4, pp. 2091–2121, 2013. DOI: 10.1109/surv.2013.032213.00009. [Online]. Available: https://www.researchgate.net/publication/256841808_Phishing_Detection_A_Literature_Survey.

[2] S. Salloum, T. Gaber, S. Vadera, and K. Shaalan, "Phishing email detection using natural language processing techniques: A literature survey," *Procedia Computer Science*, vol. 189, pp. 19–28, 2021. DOI: 10.1016/j.procs.2021.05.077. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1877050921011741.

[3] D. Irani, S. Webb, J. Giffin, and C. Pu, "Evolutionary study of phishing," *2008 eCrime Researchers Summit*, Oct. 2008. DOI: 10.1109/ecrime.2008.4696967. [Online]. Available: https://www.researchgate.net/publication/224355560_Evolutionary_study_of_phishing.

[4] O. Research, *The business cost of phishing*, Oct. 2022. [Online]. Available: https://ironscales.com/hubfs/Site%20Assets/Resources/Whitepapers/Whitepaper-Osterman-Research-The-Business-Cost-of-Phishing.pdf.

[5] R. Dhamija, J. D. Tygar, and M. Hearst, "Why phishing works," *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pp. 581–590, Apr. 2006. DOI: 10.1145/1124772.1124861. [Online]. Available: https://www.researchgate.net/publication/221516432_Why_phishing_works.

[6] A. Bergholz, J. De Beer, S. Glahn, M.-F. Moens, G. Paaß, and S. Strobel, "New filtering approaches for phishing email," *Journal of Computer Security*, vol. 18, no. 1, pp. 7–35, Jan. 2010. DOI: 10.3233/jcs-2010-0371. [Online]. Available: https://www.researchgate.net/publication/200046384_New_Filtering_Approaches_for_Phishing_Email.

[7] L. F. Gutiérrez, F. Abri, M. Armstrong, A. S. Namin, and K. S. Jones, *Phishing detection through email embeddings*, Dec. 2020. [Online]. Available: https://arxiv.org/abs/2012.14488.

[8] Y. Fang, C. Zhang, C. Huang, L. Liu, and Y. Yang, "Phishing email detection using improved rcnn model with multilevel vectors and attention mechanism," *IEEE Access*, vol. 7, pp. 56 329–56 340, 2019. DOI: 10.1109/access.2019.2913705. [Online]. Available: https://www.researchgate.net/publication/332941469_Phishing_Email_Detection_Using_Improved_RCNN_Model_With_Multilevel_Vectors_and_Attention_Mechanism.

[9] O. K. Sahingoz, 2019. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S0957417418306067.

[10] L. Eze, U. B. Chaudhry, and H. Jahankhani, "Quantum-enhanced machine learning for cybersecurity: Evaluating malicious url detection," *Electronics*, vol. 14, no. 9, p. 1827, Apr. 2025. DOI: 10.3390/electronics14091827.

[11] A. A. Aardvark, "Article title," *Journal Title*, vol. 1, no. 1, pp. 1–8, 1900.

[12] P. Sweetser, "Title of chapter in the book," in *Title of Published Book*, 1st ed., Hingham, Massachusetts, USA: Charles River Media, 2004, pp. 627–637.

[13] *Title of Published Book*, 1st ed. Hingham, Massachusetts, USA: Charles River Media, 2004.

[14] A. A. Abramson, B. B. Barbie, and C. C. Rider, "Article title," *Journal Three*, vol. 1, no. 1, pp. 192–244, 1900.

[15] J.K.Author, "Article title," *Journal Three*, vol. 1, no. 1, pp. 192–244, Jun. 1900, Accessed: June 19, 2005. doi: 10.1109/TTHZ.2016.2544142. [Online]. Available: https://ieeexplore.ieee.org/document/7463081.

[16] J. K. Author, "Title of paper," in *Abbreviated Name of Conf.*, 1900, pp. 16–19. Accessed: June 19, 2005. [Online]. Available: http://www.computer.org/csdl/proceedings/dac/2003/2394/00/2394001-abs.html.

[17] A. Basit, M. Zafar, X. Liu, A. R. Javed, Z. Jalil, and K. Kifayat, "A comprehensive survey of ai-enabled phishing attacks detection techniques," *Telecommunication Systems*, vol. 76, no. 1, pp. 139–154, Oct. 2020. DOI: 10.1007/s11235-020-00733-2. [Online]. Available: https://link.springer.com/article/10.1007/s11235-020-00733-2.

[18] P. Bountakas, K. Koutroumpouchos, and C. Xenakis, "A comparison of natural language processing and machine learning methods for phishing email detection," *Proceedings of the 16th International Conference on Availability, Reliability and Security*, pp. 1–12, Aug. 2021. DOI: 10.1145/3465481.3469205. [Online]. Available: https://www.researchgate.

net/publication/353940459_A_Comparison_of_Natural_Language_Processing_and_Machine_Learning_Methods_for_Phishing_Email_Detection.

[19] A. Junnarkar, S. Adhikari, J. Fagania, P. Chimurkar, and D. Karia, "E-mail spam classification via machine learning and natural language processing," *2021 Third International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV)*, pp. 693–699, Feb. 2021. DOI: 10.1109/icicv50876.2021.9388530. [Online]. Available: https://ieeexplore.ieee.org/document/9388530.

[20] T. Peng, I. Harris, and Y. Sawa, "Detecting phishing attacks using natural language processing and machine learning," *2018 IEEE 12th International Conference on Semantic Computing (ICSC)*, pp. 300–301, Jan. 2018. DOI: 10.1109/icsc.2018.00056. [Online]. Available: https://www.researchgate.net/publication/370701444_Detecting_Phishing_Attacks_Using_Natural_Language_Processing_And_Machine_Learning.

[21] S. Salloum, T. Gaber, S. Vadera, and K. Shaalan, "Phishing email detection using natural language processing techniques: A literature survey," *Procedia Computer Science*, vol. 189, pp. 19–28, 2021. DOI: 10.1016/j.procs.2021.05.077. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1877050921011741.

[22] M. Zareapoor and S. K. R, "Feature extraction or feature selection for text classification: A case study on phishing email detection," *International Journal of Information Engineering and Electronic Business*, vol. 7, no. 2, pp. 60–65, Mar. 2015. DOI: 10.5815/ijieeb.2015.02.08. [Online]. Available: https://www.researchgate.net/publication/276431000_Feature_

Extraction _ or _ Feature _ Selection _ for _ Text _ Classification _ A _ Case_Study_on_Phishing_Email_Detection.

[23] B. Athiwaratkun and J. W. S. W. Stokes, *Malware classification with lstm and gru language models and a character-level cnn*, 2017. [Online]. Available: https://ieeexplore.ieee.org/document/7952603/.

[24] G. Egozi and R. Verma, *Phishing email detection using robust nlp techniques | ieee conference publication | ieee xplore*, 2018. [Online]. Available: https://ieeexplore.ieee.org/document/8637476/.

[25] E. S. GUALBERTO, S. R. T. DE SOUSA, T. P. DE BRITO, and J. P. CARVALHO LUSTOSA DA COSTA, *The answer is in the text: Multi-stage methods for phishing detection based on feature engineering | ieee journals magazine | ieee xplore*, 2020. [Online]. Available: https://ieeexplore.ieee.org/document/9288667/.

[26] E. S. GUALBERTO, R. T. D. S. T. DE SOUSA, T. P. DE B. VIEIRA, J. C. L. DA COSTA, and C. G. DUQUE, *From feature engineering and topics models to enhanced prediction rates in phishing detection | ieee journals magazine | ieee xplore*, 2020. [Online]. Available: https://ieeexplore.ieee.org/document/9075252/.

[27] N. Hossain and R. Verma, *Semantic feature selection for text with application to phishing email detection*, 2013. [Online]. Available: https://www.researchgate.net/publication/312722665_Semantic_Feature_Selection_for_Text_with_Application_to_Phishing_Email_Detection.

[28] J. L. F. T. P. Y. F. A. P. H. D. M. Divakaran, 2021. [Online]. Available: https://ieeexplore.ieee.org/document/9581199/.

[29] G. Park and J. M. Taylor, *Using syntactic features for phishing detection*, May 2015. [Online]. Available: https://arxiv.org/abs/1506.00037.

[30] O. K. Sahingoz, 2019. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S0957417418306067.

[31] R. Valecha, P. Mandaokar, and H. R. Rao, *Phishing email detection using persuasion cues | ieee journals magazine | ieee xplore*, 2021. [Online]. Available: https://ieeexplore.ieee.org/document/9565347/.

[32] D. Wagner, C. Karlof, J. Tygar, and U. Shankar, *Dynamic pharming attacks and locked same-origin policies for web browsers*, 2007. [Online]. Available: https://dl.acm.org/doi/10.1145/1315245.1315254.

[33] OpenAI, *Chatgpt (mar 2024 version)*, Used for summarizing journal articles and literature synthesis, San Francisco, CA, 2024. [Online]. Available: https://chat.openai.com.

[34] Grammarly, Inc., *Grammarly*, Used for grammatical correction and stylistic editing, San Francisco, CA, 2024. [Online]. Available: https://www.grammarly.com.